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#include <xc.h>
#include <stdint.h>
#include "config.h"
#define _XTAL_FREQ 4000000
//--[ Pin Definitions ]--
#define Rev RB0 // Reverse Direction
#define LV0 RB1 // 0% Speed Button
#define LV1 RB2 // 50% Speed Button
#define LV2 RB3 // 75% Speed Button
#define LV3 RB4 // 100% Speed Button
//=====
//--[ Function Declarations ]--
void PWM1_Set_Duty(uint16_t);

void main(void)
{
    //--[ Configuration For IO Pins ]--
    TRISB = 0x1F; // Low 5-pins are inputs
    // Set The Direction To Be Output Pins
    TRISD0 = 0;
    TRISD1 = 0;
    // Initially (0, 1) Say it's ClockWise
    // For Reversing The Direction Write
    RD0 = 0;
    RD1 = 1;

    CCP1M3 = 1;
    CCP1M2 = 1;
    TRISC2 = 0; // The CCP1 Output Pin (PWM)
    // Set The PWM Frequency (2kHz)
    PR2 = 124;
    // Set The PS For Timer2 (1:4 Ratio)
    T2CKPS0 = 1;
    T2CKPS1 = 0;
    // Start CCP1 PWM !
    TMR2ON = 1;

    // The Main Loop (Routine)
    while(1)
    {
        if(Rev) // Reverse The Direction
        {
            RD0 = ~RD0;
            RD1 = ~RD1;
            __delay_ms(500); // Wait For The Button To Be Released
        }
        if(LV0) // 0% DC
        {
            PWM1_Set_Duty(0);
            __delay_ms(100); // Wait For The Button To Be Released
        }
    }
}

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    }
    if(LV1) // 50% DC
    {
        PWM1_Set_Duty(250);
        __delay_ms(100); // Wait 100ms
    }
    if(LV2) // 75% DC
    {
        PWM1_Set_Duty(375);
        __delay_ms(100); // Wait 100ms
    }
    if (LV3) // 100% DC
    {
        PWM1_Set_Duty(500);
        __delay_ms(100); // Wait 100ms
    }
    __delay_ms(10); // Wait To Reduce The Simulation Overloading
}
return;
}

```

```

// Definition For PWM1_Set_Duty Function
void PWM1_Set_Duty(uint16_t DC)
{
    // Check The DC Value To Make Sure it's Within 10-Bit Range
    if(DC<1024)
    {
        CCP1Y = DC & 1;
        CCP1X = DC & 2;
        CCPR1L = DC >> 2;
    }
}

```